

MAYANK TELI

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EDUCATION

National Institute of Technology, Tiruchirappalli (NIT Trichy)

Jul 2019 - May 2023

Bachelor of Technology in Instrumentation and Control Engineering

CGPA: 8.1 / 10

WORK EXPERIENCE

Samsung R&D Institute, Bangalore

Nov 2023 - Present

Machine Learning Engineer, Home AI and Domain Agents

- Built a production-grade multimodal device troubleshooting assistant combining LLM-based planning (Gemini Live), **RAG** over device manuals, and YOLO-based visual perception, enabling real-time guidance.
- Designed a two-stage visual state detection pipeline with **YOLO**-based device/component detection (mAP@50: 0.92) followed by ROI-based classifiers and geometric validation rules, improving procedural step verification accuracy by 28 % across 15+ maintenance workflows.
- Built a transformer-based multi-head NER model for Samsung Gallery and applied **LoRA** fine-tuning, achieving 96–97% token-level accuracy while reducing inference latency by 70–80% compared to a fully fine-tuned baseline.
- Reproduced and benchmarked state-of-the-art research models – **iSEARLE** (Recall@10: 76%) and **CLIP4CIR** (Recall@10: 80%) on CIRR Dataset for **Multimodal Search** in gallery

Samsung R&D Institute, Bangalore

Jun 2022 – Aug 2022

Magpie Intern – SmartThings Platform

- Integrated and controlled 10+ smart devices via Open APIs on SmartThings App.
- Validated app functionality across devices ensuring interoperability and performance.

PATENT

Multimodal Image Retrieval Using Image-Conditioned Text Queries Co-Inventor — Patent Filed (**A1 Grade**, HQ Approved)—2025

PROJECTS

CLIP from Scratch: Contrastive Language–Image Pretraining — [GitHub](#)

2025

- Implemented a **CLIP-style multimodal model** from scratch, jointly training image and text encoders using contrastive loss.
- Studied the impact of batch size, temperature scaling, and negative sampling on image–text alignment.
- Evaluated image–text retrieval performance and analyzed embedding similarity behavior.

Vision Transformer from Scratch — [GitHub](#)

2025

- Implemented and trained a Vision Transformer from scratch, experimenting with architecture depth, patch size, and attention heads.
- Conducted experiments across datasets to analyze convergence behavior and performance trade-offs.

TECHNICAL SKILLS

Languages	Python, Java
Frameworks	PyTorch, TensorFlow, Flask, FastAPI, Optuna, MLflow
Domains	NLP, Computer Vision, LLMs, Generative AI, RAG
Libraries	NumPy, Pandas, scikit-learn, Matplotlib, Hugging Face Transformers, OpenCV

ACHIEVEMENTS

- Codeforces Peak Rating: 1501 (Specialist) – [mayank61](#)
- Leetcode Peak Rating: 1907 – [mayankteli1309](#)
- Solved over 2700+ coding problems across platforms